

8.17.16.2 Pressure ratio data cards, CCC7NN1-NN9

NN can be 01 to 10. This number must be equal to the W1(I) of CCC70000 card.

W1(R) Enter the RPM number for the pressure ratio data set.

W2(R) Enter the corrected mass flow rate.

W3(R) Enter the pressure ratio.

W4(R)~W41(R) Additional corrected mass flow rate and pressure ratio pairs. Up to 20 pairs can be entered for one RPM set.

8.17.17 Tabulated circulator efficiency data input, CCC8NNX

These cards are required for circulator. Efficiency data are tabulated by RPM and corrected mass flow rate. The number NN may be 01-10. The quantity X can be 1 to 9.

8.17.17.1 Number of data card, CCC8000

Enter the number of tabulated efficiency data sets. A data set is the corrected mass flow rate and efficiency pairs for a RPM value.

W1(I) NumEffTBL. Number of efficiency data sets.

8.17.17.2 Efficiency data cards, CCC8NN1-NN9

NN can be 01 to 10. This number must be equal to the W1(I) of CCC80000 card.

W1(R) Enter the RPM number for the efficiency data set.

W2(R) Enter the corrected mass flow rate.

W3(R) Enter the efficiency.

W4(R)~W41(R) Additional corrected mass flow rate and efficiency pairs. Up to 20 pairs can be entered for one RPM set.

8.18 Multiple Junction Component, MTPLJUN

A multiple junction component is indicated by MTPLJUN for Word 2 on card CCC0000.

The one or more junctions specified by this component can connect volumes in the same manner as several single-junction components except that all the volumes connected by the junctions in the

component must be in the same hydrodynamic system. If this restriction is violated, corrective action is to merge the hydrodynamic systems. For major edits, minor edits, and plot variables, the junctions in the multiple junction component are numbered CCCJJNN00, where NN is the set number and JJ is the junction number within the set. The quantity NN may be 01-99; JJ is 01 for the first junction described in a set and incremented by 1 for each additional junction ($01 \leq JJ \leq 99$).

8.18.1 Multiple Junction Information, CCC0001

This card is required.

- W1(I) Number of junctions, nj. This number must be > 0 and < 100 .
- W2(I) Initial condition control. This word is optional and, if missing, is assumed to be 0. If 0 is entered, the initial conditions on cards CCC1NNM are velocities; if 1 is entered, the initial conditions are mass flows.

8.18.2 Multiple Junction Geometry, CCC0NNM

These cards are required. Junctions are described by one or more sets of data, NN being the set number and M being the card number within a set. The junctions are numbered as CCCJJNN00, where JJ is 01 for the first junction described in a set and increments by 1 for each additional junction. The quantity NN may be 01-99, and M may be 1-9. cards are processed by increasing set number NN, and cards within a set by increasing M. Neither NN or M need be strictly consecutive.

- W1(I) From connection code to a component. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction. The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.
- W2(I) To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.
- W3(R) Area (m^2 , ft^2). If 0, the area is set to the minimum area of the adjoining volumes. For abrupt area changes, the area must be equal to or smaller than the minimum of the adjoining volume areas. For smooth area changes, there are no restrictions.
- W4(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section 8.18.5).

The interpretation and use of this loss coefficient vary based upon the “a” junction control flag selected in W6. If the smooth area change option is selected ($a = 0$), the entry here is the loss coefficient appropriate for the junction flow area resulting from the entry made in W3. If one of the abrupt area change options is selected ($a = 1$ or 2), the entry here is the loss coefficient appropriate for the minimum flow area between the two hydrodynamic volumes connected by this junction. If the $a = 1$ option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

- W5(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. A variable loss coefficient may be specified (see Section 8.18.5). See information for the previous word regarding interpretation and use of this loss coefficient.
- W6(I) Junction control flags. This word has the format jefvcahs. It is not necessary to input leading zeros. For the multiple junction, the junction control flag is limited to the format 0ef0cahs.
- e-flag The digit e specifies the modified PV term in the energy equations. $\underline{e} = 0$ means that the modified PV term will not be applied, and $\underline{e} = 1$ means that it will be applied.
- f-flag The digit f specifies CCFL options. $\underline{f} = 0$ means that the CCFL model will not be applied, and $\underline{f} = 1$ means that the CCFL model will be applied.
- c-flag The digit c specifies choking options. $\underline{c} = 0$ means that the choking model will be applied, and $\underline{c} = 1$ means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.
- a-flag The digit a specifies area change options. $\underline{a} = 0$ means a smooth area change and only the user supplied K_{loss} is applied. $\underline{a} = 1$ means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. $\underline{a} = 2$ means a partial abrupt area change model and it is the same as “ $\underline{a} = 1$ ” except no code calculated forward and reverse contraction/expansion K_{loss} is applied.
- h-flag The digit h specifies nonhomogeneous or homogeneous. $\underline{h} = 0$ specifies the nonhomogeneous (two-velocity momentum equations) option; $\underline{h} = 1$ or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option ($\underline{h} = 1$ or 2), the major edit printout will show $\underline{h} = 1$.

- s-flag The digit s specifies momentum flux options. s = 0 uses momentum flux in both the to and from volume. s = 1 uses momentum flux in the from volume, but not in the to volume. s = 2 uses momentum flux in the to volume, but not in the from volume. s = 3 does not use momentum flux in either the to volume or the from volume.
- W7(R) The meaning of this word is different, depending on the critical flow model which has been selected (see W6).
- If the Henry-Fauske critical flow model is active, the discharge coefficient is entered. The recommended default value for this coefficient is 1.0.
- If the original RELAP5 critical flow model is active, the subcooled discharge coefficient is entered. This entry applies only to subcooled liquid choked flow calculations. The entry must be > 0.0 and < 2.0. The recommended default value for this coefficient is 1.0.
- W8(R) The meaning of this word is different, depending on the critical flow model which has been selected (see W6).
- If the Henry-Fauske critical flow model is active, the thermal nonequilibrium constant is entered. The recommended default value for this coefficient is 0.14. If the entry is < 0.01, the equilibrium option is used and the entry is reset to 0.0. If the entry is > 1000.0, the frozen option is used and the entry is reset to 100.0.
- If the original RELAP5 critical flow model is active, the two-phase discharge coefficient is entered. This entry applies only to two-phase choked flow calculations. The entry must be > 0.0 and < 2.0. The recommended default value for this coefficient is 1.0.
- W9(R) The meaning of this word is different, depending on the critical flow model which has been selected (see W6).
- If the Henry-Fauske critical flow model is active, enter 0.0 (this word is not used).
- If the original RELAP5 critical flow model is active, the superheated discharge coefficient is entered. This entry applies only to superheated vapor choked flow calculations. The entry must be > 0.0 and < 2.0. The recommended default value for this coefficient is 1.0.
- W10(I) From volume increment. Words 1 and 2 contain the from and to connection codes respectively for the first junction defined by the set. If the set defines more than one junction, connection codes for the following junctions are given by the connection code of the previous junction plus the increments in Words 10 and 11. The increments may be positive, negative, or 0. Junctions are defined up to the limit in Word 13. Words 3-9 apply to all junctions defined by the set. If additional sets are entered, Words 1 and 2 apply to the next junction, and increments are applied as with the first set. Word 13 for the second and

following sets must be greater than Word 13 of the preceding set, and Word 13 of the last set must equal n_j . A new set is used whenever a new increment is needed, Words 3-9 need to be changed, or a change in junction numbering is desired.

W11(I) To volume increment. See description for Word 10.

W12(I) Enter 0. This is reserved for future capability.

W13(I) Junction limit. Described above.

8.18.3 Multiple Junction Initial Condition, CCC1NNM

These cards are required. Initial velocities are entered using one or more sets of data. The processing of sets of data is identical to that described in Section 8.18.2 except that there need be no relationship in the division of junctions within sets between these cards (CCC1NNM) and the multiple junction geometry cards (CCC0NNM). Likewise, these cards do not affect the numbering of the junctions.

W1(R) Initial liquid velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s), depending on control Word 2 of card CCC0001.

W2(R) Initial vapor velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s), depending on control Word 2 of card CCC0001.

W3(I) Junction limit number.

8.18.4 Multiple Junction Diameter and CCFL Data, CCC2NNM

These cards are optional. The defaults indicated for each word are used if the card is not entered. If the card is being used to specify only the junction hydraulic diameter for the interphase drag calculation (i.e., $f = 0$ in Word 6 of cards CCC0NNM), then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used). If this card is being used for the CCFL model (i.e., $f = 1$ in Word 6 of cards CCC0NNM), then enter all four words for the appropriate CCFL model if values different from the default values are desired. The processing of sets of data is identical to that described in Section 8.18.2 except that there need be no relationship in the division of junctions within sets between these cards (CCC1NNM) and the multiple junction geometry cards (CCC0NNM). Likewise, these cards do not affect the numbering of the junctions.

W1(R) Junction hydraulic diameter, D_j (m, ft). This is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag and must be ≥ 0 . This number should be computed from $4.0 \cdot \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If a 0 is entered or if the default is used,

the junction diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi} \right)^{0.5}$. See Word 3 of card CCC0NNM for area.

- W2(R) Flooding correlation form, β . If 0, the Wallis CCFL form is used. If 1.0, the Kutateladze CCFL form is used. If between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be ≥ 0 and ≤ 1 . The default value is 0 (Wallis form).
- W3(R) Gas intercept, c . This is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0 . The default value is 1.
- W4(R) Slope, m . This is the slope used in the CCFL correlation and must be > 0 . The default value is 1.
- W5(I) Junction limit number.

8.18.5 Multiple Junction Form Loss Data, CCC3NNM

These cards are optional. The user-specified form loss is given in Words 4 and 5 of card CCC0NNM if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F}$$

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 4 and 5 of card CCC0NNM. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all five words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

- W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.
- W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.
- W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.
- W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.
- W5(I) Junction limit number.